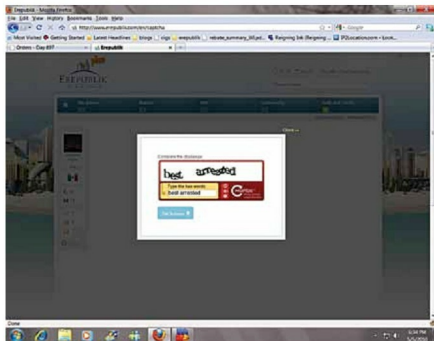


An Insight into CAPTCHA

You may hate those blurry letters, putting you off filling in an online form. But have you ever thought what exactly are these simple tests for? Here is an introduction



Encounter with CAPTCHA

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While surfing the Internet, many a times you need to create accounts on different Web domains and fill in some online forms. During registration process or on entering the wrong password while logging into an account, you encounter a code containing blurry or distorted letters. You are prompted to enter these distorted characters in a textbox. The registration or login is allowed only if you correctly enter this code in the textbox.

To most of us, this requirement appears just as a hindrance. But have you ever thought what exactly are these simple tests for, and is there any significance of their use?

To give you an introduction, this is actually one of the Web testing mechanisms referred to as CAPTCHA. The term CAPTCHA, an acronym for 'Completely Automated Public Turing Test to Tell Computers and Humans Apart,' was coined in 2000 by Louis von Ahn, Nicholas Hopper and John Langford of Carnegie Mellon University.

CAPTCHA has its foundation in an experiment called Turing test. The Turing test is a test of a machine's ability to exhibit intelligent behaviour. The classic Turing test is a game of imitation comprising an interrogator and two participants, of which one is a human and the other a machine (see Fig. 1). The interrogator sits in one room and the participants are made to sit in different rooms. The setup is such

that the participants are not visible to the interrogator.

The interrogator asks both the participants some questions and from their responses tries to distinguish between the human and the machine. If the interrogator is unable to distinguish, the machine passes the Turing test.

CAPTCHA is basically a type of challenge-response test used in computing as an attempt to ensure that the response is generated by a human. The tests are designed such that humans could easily pass them but not the machine. Thus the test identifies which users are actually real human beings and which are automated programs. It is also referred to as reverse Turing test as it is administered by a machine and targeted at humans.

This raises the question "why the CAPTCHA was introduced?" The reason is that there may be someone

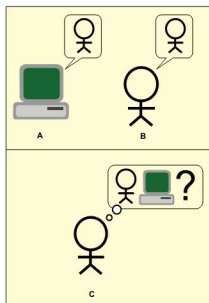


Fig. 1: Turing test—the basis of CAPTCHA